

Ren Wang

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Education

National Taiwan University , Taipei, Taiwan <i>Ph.D. Candidate in Computer Science and Information Engineering</i> Advisor: Yung-Yu Chuang	2021–present
National Tsing Hua University , Hsinchu, Taiwan <i>M.S. in Computer Science</i> Advisor: Hwann-Tzong Chen	2013–2015
National Tsing Hua University , Hsinchu, Taiwan <i>B.S. in Computer Science</i>	2009–2013

Work Experience

Google Taiwan , New Taipei City, Taiwan <i>Senior Software Engineer, Pixel Camera</i>	10/2024–present
MediaTek Inc. , Hsinchu, Taiwan <i>Senior Engineer, Multimedia Development Div. XVI</i> <i>Senior Engineer, Multimedia Technology Development Div.</i> <i>Engineer, Multimedia Technology Development Div.</i> <i>Engineer, Advanced Communication Technology Div.</i>	03/2023–06/2024 06/2020–02/2023 02/2018–05/2020 12/2015–01/2018

Selected Projects

Unified Codebase for AI-ISP Algorithm Development , MediaTek	2023–2024
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ACHIEVEMENTS

- Integrated 4 AI-ISP projects into a unified codebase with colleagues, where the projects included image denoising, video denoising, image enhancement, and remosaicing.
- Enabled 10+ engineers to utilize this codebase for developing algorithms within AI-ISP projects.

RESPONSIBILITIES

- Led a team of 5+ engineers in the construction of the codebase's infrastructure.
- Established collaboration guidelines for cross-project and cross-functional teams.
- Improved the software architecture to support the diverse needs of AI-ISP projects.

AI-Noise Reduction in Low-Light Captures , MediaTek 	2017–2022
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ACHIEVEMENTS

- Contributed to scaling this project across 25 smartphone series from June 2019 to Feb 2022, where the clients included OPPO, vivo, Xiaomi, realme, and OnePlus.

RESPONSIBILITIES

- Applied cutting-edge AI techniques to image denoising on MediaTek's image signal processors.
- Optimized neural networks to reduce the computation time and memory/power consumption on MediaTek's deep learning accelerators.
- Collaborated closely with relevant projects, such as image enhancement, demosaicing, and HDR.

Activity and Gesture Recognition , MediaTek	2016–2017
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ACHIEVEMENTS

- Contributed to deploying this project on Helio X30 in 2017, where the client was Meizhu.

RESPONSIBILITIES

- Implemented statistical models and optimized their computational costs on mobile platforms, where the models included SVM, GMM, and HMM.
- Prototyped deep learning models for gesture recognition using IMU sensors.

Publications

- **Ren Wang** and Yung-Yu Chuang, “Uncertainty-guided latent diffusion models for faithful super resolution,” in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, Sept 2026.
- Shao-Hao Lu, **Ren Wang**, Ching-Chun Huang, and Wei-Chen Chiu, “Boosting diffusion guidance via learning degradation-aware models for blind super resolution,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Feb 2025. **[Oral Presentation]**
- **Ren Wang**, Yu-Lun Liu, Yu-Hao Huang, and Ning-Hsu Wang, “Methods and apparatuses of depth estimation from focus information,” U.S. Patent 11,967,096, issued Apr 23, 2024.
- Ning-Hsu Wang, **Ren Wang**, Yu-Lun Liu, Yu-Hao Huang, Yu-Lin Chang, Chia-Ping Chen, and Kevin Jou, “Bridging unsupervised and supervised depth from focus via all-in-focus supervision,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct 2021.
- Chien-Chuan Su, **Ren Wang**, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Soo-Chang Pei, “Explorable tone mapping operators,” in *Proceedings of the International Conference on Pattern Recognition (ICPR)*, Jan 2021.
- Ke-Chi Chang, **Ren Wang**, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Hwann-Tzong Chen, “Learning camera-aware noise models,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, Aug 2020.

Honors

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| • Google Spot Bonus (for achievements recognized by managers) | 2025 |
| • MediaTek vAward (16 times; for exemplary performance) | 2015–2024 |
| • MediaTek Project Award (for Dimensity 1000) | July 2020 |
| • Valedictorian , NTHU CS | June 2013 |
| • Finalist of the Senior Project Contest , NTHU CS | Oct 2012 |

Academic Activities

Paper Reviewer

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| • IEEE Transactions on Image Processing | |
| • Conference on Neural Information Processing Systems (NeurIPS) | 2026 |
| • European Conference on Computer Vision (ECCV) | 2026 |
| • IEEE International Conference on Image Processing (ICIP) | 2026 |
| • IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) | 2026 |
| • IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) | 2025 |

Teaching Assistant, National Tsing Hua University

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| • ISA 525700: Computer Vision for Visual Effects | Spring 2015 |
| • EECS 111000: Introduction to Programming (C language) | Fall 2013, 2014 |
| • CS 321100: Introduction to Communication | Fall 2014 |

Technical Skills

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| Expertise | Generative AI, Deep Learning, Computational Photography, Low-Level Vision |
| Programming | Python, C/C++, Git, PyTorch, JAX (Flax/Optax), TensorFlow, scikit-learn, OpenCV |